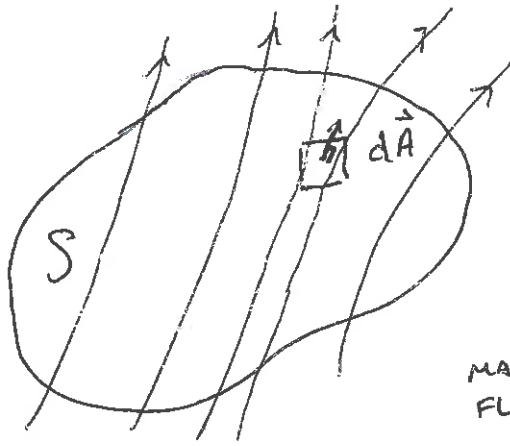




$\vec{B}(t)$

$$\Phi_B = \int_S \vec{B}(t) \cdot d\vec{A}$$

MAGNETIC FLUX

MAGNETIC FLUX DENSITY

$$\frac{d\Phi_B}{dt} = \frac{d}{dt} \int_S \vec{B}(t) \cdot d\vec{A}$$

$$\frac{d\Phi_B}{dt} = \int_S \left(\frac{d}{dt} \vec{B}(t) \right) \cdot d\vec{A} + \int_S \vec{B}(t) \cdot \frac{d\vec{A}}{dt}$$

FARADAY'S LAW

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int_S \vec{B}(t) \cdot d\vec{A}$$

INTEGRAL AROUND
CLOSURE / WIRE ETC.

$$\oint \vec{E} \cdot d\vec{l} = \frac{d\Phi_B}{dt}$$

ELECTRIC FIELD

MAGNETIC
TIME-VARYING FLUX

(TIME-VARYING
MAGNETIC FIELD)

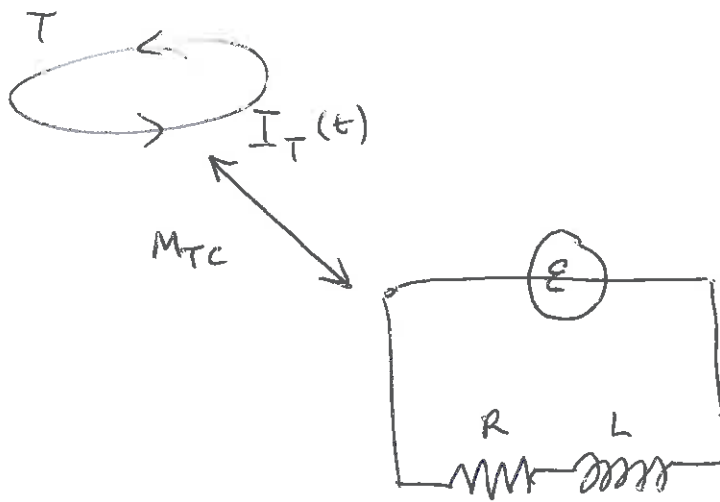
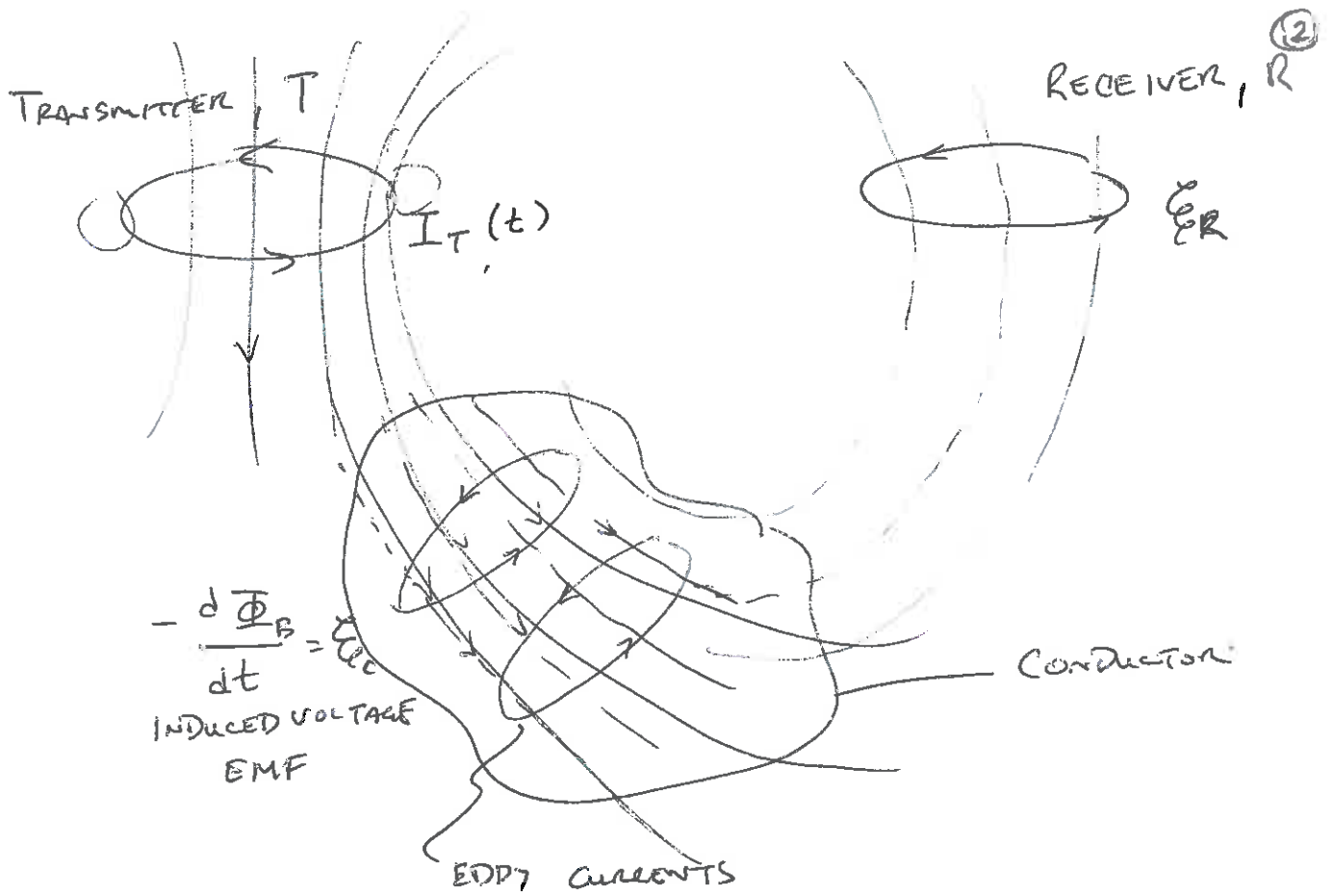
LORENTZ FORCE LAW

$$\vec{F} = q [\vec{E} + \vec{v} \times \vec{B}]$$

$$\vec{F} = q \vec{E} \Rightarrow \vec{E} = \frac{\vec{F}}{q}$$

$$\mathcal{E} = \text{ELECTROMOTIVE FORCE} = \oint_C \frac{\vec{F} \cdot d\vec{l}}{q} \quad \frac{[N][m]}{[C]} = [V]$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = - \frac{d\Phi_B}{dt}$$



KIRKOFF'S LOOP LAW

$$\epsilon - V_L - V_R = 0$$

$$\epsilon = -M_{TC} \frac{dI_T(t)}{dt}$$

DRIVING VOLTAGE

$$V_L = L \frac{dI_c(t)}{dt}$$

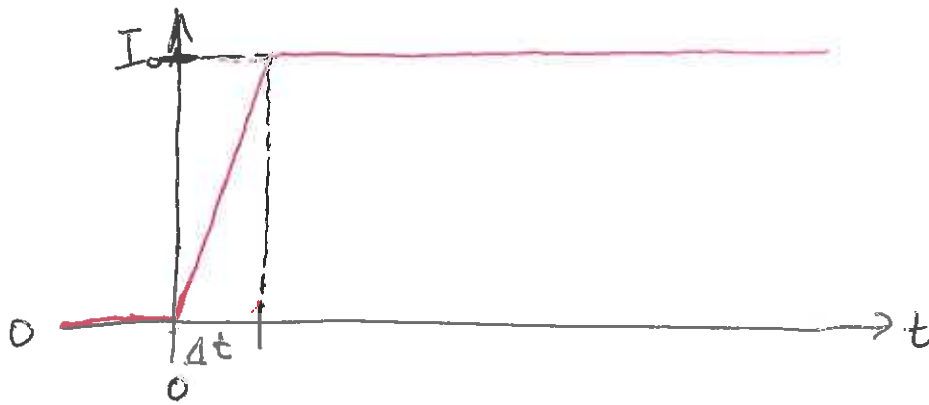
RETARDING

$$V_R = RI_c(t)$$

DISSIPATION

I_T UNDER OUR CONTROL !!!

③



$$I_T(t) = I_0 u(t)$$

↑
RAMP FUNCTION

$$u(t) = \begin{cases} 0 & t < 0 \\ t/\Delta t & 0 \leq t \leq \Delta t \\ 1 & t > \Delta t \end{cases}$$

$$\begin{aligned} \mathcal{E} &= -M_{TC} \frac{dI_T(t)}{dt} \\ &= -M_{TC} \frac{d(I_0 u(t))}{dt} = -M_{TC} I_0 \frac{du(t)}{dt} \end{aligned}$$

$$\mathcal{E} = V_L + V_R$$

$$-M_{TC} I_0 \frac{du(t)}{dt} = L \frac{dI_c(t)}{dt} + R I_c(t)$$

$$\frac{du(t)}{dt} = \begin{cases} \frac{d(0)}{dt} & t < 0 \\ \frac{d(t/\Delta t)}{dt} & 0 \leq t \leq \Delta t \\ \frac{d(1)}{dt} & t > \Delta t \end{cases} = \begin{cases} 0 & t < 0 \\ \frac{1}{\Delta t} & 0 \leq t \leq \Delta t \\ 0 & t > \Delta t \end{cases}$$

$$1: t < 0 \quad \frac{d u(t)}{dt} = 0$$

(4)

$$0 = L \frac{d I_c(t)}{dt} + R I_c(t)$$

$$\boxed{I_c = 0 \quad t < 0}$$

$$2: 0 \leq t < \Delta t, \quad \frac{d u(t)}{dt} = \frac{1}{\Delta t}$$

$$-\frac{M_{Tc} I_0}{\Delta t} = L \frac{d I_c(t)}{dt} + R I_c(t)$$

$$c_2 = c_0 \frac{dy}{dt} + c_1 \frac{dy}{dt}$$

$$y = y_0 \exp\left(-t \frac{c_1}{c_0}\right) + \frac{c_2}{c_0}$$

$$y_0 \stackrel{!}{=} I_c, \quad t = 0, \quad y = 0$$

$$y(0) = 0 = y_0 \exp\left(-0 \frac{c_1}{c_2}\right) + \frac{c_2}{c_0}$$

$$\frac{c_2}{c_0} = y_0$$

$$I_c(t) = -\frac{M_{Tc} I_0}{L \Delta t} \left[1 - \exp\left(-t \frac{R}{L}\right) \right]$$

$$t \ll \frac{L}{R}$$

$$\exp(x) = (-x + \frac{1}{2}x^2 - \dots)$$

$$I_c(t) = -\frac{M_{Tc} I_0}{L \Delta t} t$$

$$I_c(t = \Delta t) = -\frac{M_{Tc} I_0}{L} \quad 0 \leq t \leq \Delta t$$

3: $t > \Delta t$, $\frac{d u(t)}{dt} = 0$

5

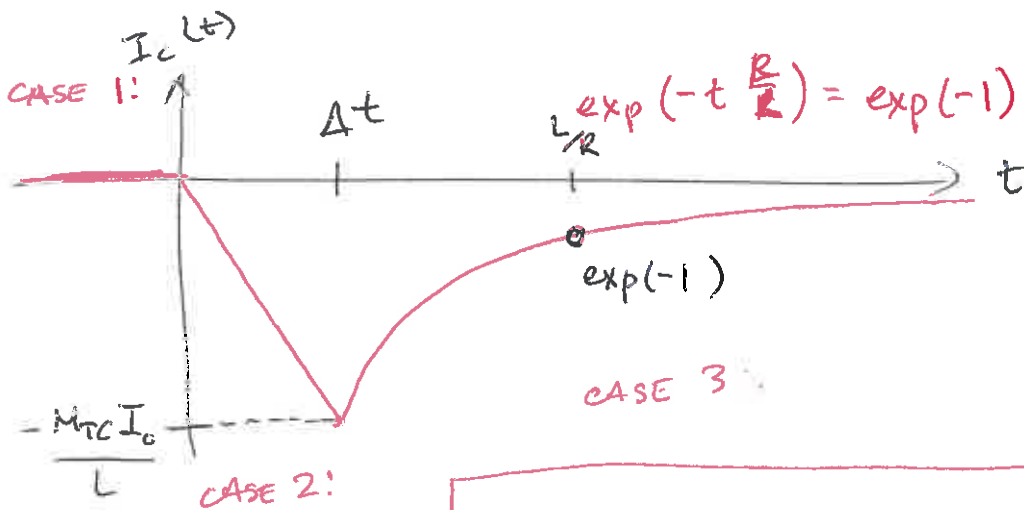
$$0 = \underbrace{L}_{a_0} \frac{d I_c(t)}{dt} + \underbrace{R}_{a_1} I_c(t)$$

$$0 = a_0 \frac{dy}{dt} + a_1 y$$

$$y = K \exp\left(-t \frac{a_1}{a_0}\right)$$

$$y(0) = I_c(\Delta t) = - \frac{M_{TC} I_0}{L} = K$$

$$I_c(t) = - \frac{M_{TC} I_0}{L} \exp\left(-t \frac{R}{L}\right) \quad t > \Delta t$$



$R \gg L$ RAPID DECAY = RESISTOR

$R \ll L$ SLOW DECAY = CONDUCTOR